

Reassessing Regional Favoritism in Ethiopia: A Descriptive Analysis Using Nighttime Lights and GDP Data

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March 20, 2026

Abstract

This paper provides a transparent, descriptive reassessment of regional economic inequality and alleged political favoritism in Ethiopia since the introduction of ethnic federalism. Leveraging harmonized nighttime lights (NTL) and GDP data from 1992 to 2024, we document trends in regional economic activity and test for associations between national leaders' birth regions and regional development outcomes. In line with best practices for transparency and reporting standards in applied economics [Christensen and Miguel, 2018, Nosek et al., 2015], we explicitly report minimum detectable effect sizes and statistical power [Gerber et al., 2010]. While NTL strongly proxies for GDP at the national level [Chen and Nordhaus, 2011, Henderson et al., 2012, Li and Gibson, 2023], and regional inequality has declined, we find no robust evidence that leaders' birth regions systematically outperform others. However, the minimum detectable effect size in this dataset is nearly four times larger than typical effects reported in the literature; only extremely large favoritism effects can be ruled out. Methodological challenges—including a small number of regions, infrequent leadership transitions, and measurement limitations—preclude credible causal inference and limit statistical power [Angrist and Pischke, 2009, Marianne Bertrand and Mullainathan, 2004, Imbens and Rubin, 2015]. Our findings highlight the need for richer data, more exogenous variation, and careful diagnostics in future research on distributive politics and regional inequality [Bickenbach and Bode, 2016, Bank, 2024].

1 Introduction

The distributive consequences of political power remain a central concern in development economics and political science [Persson and Tabellini, 2000]. A growing literature finds that national leaders often channel public resources and economic opportunities toward their birth regions, a phenomenon widely described as regional favoritism [Hodler and Raschky, 2014, Beiser-McGrath and Bernauer, 2021, Onana et al., 2024]. Much of this

evidence relies on satellite-based proxies for economic activity, particularly harmonized nighttime lights (NTL) [Chen and Nordhaus, 2011, Henderson et al., 2012, Li and Gibson, 2023, Gibson and Li, 2021], combined with panel data methods.

However, the empirical foundations of these findings remain uncertain. Many studies rely on settings with limited within-unit variation, infrequent leadership transitions, and potential collinearity between treatment and time effects. In such contexts, standard panel estimators may be unstable or highly model-dependent, especially in small-N settings; this raises concerns about the robustness and interpretability of reported favoritism effects. While Marianne Bertrand and Mullainathan [2004], Angrist and Pischke [2009], Imbens and Rubin [2015] discuss inference and identification challenges, simulation evidence and recent reviews highlight the risks of model dependence and instability when identifying variation is limited.

Ethiopia provides a stringent test case for these concerns. The country combines centralized fiscal authority, harmonized administrative boundaries, and a small number of regions with relatively infrequent leadership transitions. These features severely constrain the identifying variation required to estimate within-region effects. If regional favoritism is present and detectable with the available data and methods, it should appear in this context. Conversely, failure to detect consistent effects may reflect limitations of the empirical design rather than the absence of favoritism.

This paper asks whether national leaders’ birth regions in Ethiopia experience systematically higher economic performance, as measured by harmonized nighttime lights and regional GDP, relative to other regions. Answering this question is empirically challenging due to limited within-region variation, high collinearity between treatment and time effects, and potential measurement error in both NTL and GDP.

This paper makes three contributions. First, it offers a descriptive analysis of regional economic dynamics in Ethiopia using harmonized NTL and GDP data from 1992 to 2024. Second, it shows that in small-N panel settings with few treatment switches, standard panel estimators can yield unstable and specification-dependent estimates, with statistical significance that does not persist across alternative models or diagnostic tests. Third, the paper quantifies the limits of inference in this setting by showing that only very large effects—substantially larger than those typically reported in the literature—are detectable given the available data and variation.

Taken together, these findings imply that the absence of robust evidence should be interpreted cautiously: the empirical design lacks the power and identifying variation required to detect effects of economically relevant magnitudes. More broadly, the results highlight the need for greater transparency in reporting statistical power and design limitations, and suggest that credible inference in this literature requires either richer data or exogenous sources of variation.

To address these questions, we assemble a panel of harmonized NTL and GDP data for Ethiopian regions from 1992 to 2024 and apply fixed-effects models alongside placebo and event-study diagnostics. While some specifications suggest leader-region advantages, these estimates are not robust across alternative specifications or identification tests. The overall pattern indicates that standard empirical strategies provide limited and fragile evidence in this context.

2 Literature Review

This study relates to three strands of literature: the use of nighttime lights (NTL) as a proxy for economic activity, research on regional favoritism in political economy, and work on federalism and regional development in Ethiopia.

First, a large and growing literature uses satellite-based luminosity to measure economic activity, particularly in data-constrained environments. Early contributions show that nighttime lights correlate strongly with national and regional output, making them a valuable proxy where conventional statistics are limited [Chen and Nordhaus, 2011, Henderson et al., 2012]. More recent work, however, highlights important limitations. The strength of the relationship between lights and economic activity varies with population density, sectoral composition, and spatial scale, complicating subnational interpretation [Bluhm and Storeygard, 2022, Gibson and Li, 2021]. Advances in harmonized long-run datasets have improved comparability across satellite systems, but measurement challenges remain, particularly for local-level analysis [Li and Gibson, 2023, Gibson, 2024]. These findings suggest that while NTL data are informative, their interpretation depends on context and level of aggregation. This is particularly relevant for studies of regional favoritism, where researchers often rely on NTL as a proxy for subnational economic outcomes.

Second, this paper connects to the literature on regional and ethnic favoritism. Cross-country evidence suggests that national leaders' birth regions tend to experience higher levels of economic activity, as proxied by nighttime lights [Hodler and Raschky, 2014]. Subsequent studies emphasize the role of political institutions, ethnic demography, and local power structures in shaping the distribution of resources [Beiser-McGrath and Bernauer, 2021]. Related work documents geographically targeted allocation of public goods, investment, and clientelist transfers across a range of settings [Golden and Min, 2013, Onana et al., 2024, Cruzatti et al., 2024]. While these findings point to the importance of political favoritism, some studies note that panel data designs can face challenges in settings with limited within-unit variation and infrequent leadership transitions [Marianne Bertrand and Mullainathan, 2004, Angrist and Pischke, 2009, Imbens and Rubin, 2015]. These methodological concerns are especially salient in the Ethiopian context, where the structure of federalism and the distribution of political power shape both the opportunities for favoritism and the feasibility of empirical identification.

Third, the paper builds on research on federalism and regional development in Ethiopia. Although Ethiopia formally operates under a federal system, prior work highlights the persistence of centralized control over fiscal policy, investment decisions, and administrative coordination [Faguet, 2021, Wubneh, 2017]. Theoretically, centralization limits the autonomy of regional governments, reducing their ability to independently allocate resources or pursue region-specific development strategies. This mechanism constrains both the scope for distributive favoritism and the channels through which it might operate. Empirically, this means that even if favoritism exists, its observable impact may be muted or difficult to detect in standard panel designs, as most variation in resource allocation is determined at the center rather than the region. At the same time, recent political developments have introduced greater regional contestation and variation in local governance, potentially altering patterns of resource allocation and complicating identification.

Taken together, these literatures highlight both the relevance and the empirical difficulty of identifying regional favoritism—especially in contexts like Ethiopia, where the

combination of centralized authority, limited regional variation, and evolving political institutions present unique challenges for both measurement and inference.

Despite advances in measuring economic activity with nighttime lights and a growing body of evidence on regional favoritism, there remains a critical gap in understanding how well standard empirical strategies perform in contexts with limited within-unit variation, infrequent leadership transitions, and centralized political authority. Most existing studies implicitly assume that panel data methods are robust across settings, yet these assumptions may not hold in countries like Ethiopia, where the institutional and data environment sharply constrain identification. This study directly addresses this gap by asking: Can standard empirical approaches reliably detect regional favoritism in Ethiopia, or do data and design limitations fundamentally undermine inference in this context?

3 Empirical Strategy and Methods

3.1 Empirical Strategy

This study uses a panel of Ethiopian regions from 1992 to 2024 to examine whether national leaders' birth regions experience differential economic performance, as measured by harmonized nighttime lights (NTL) and, where available, regional GDP [Li and Gibson, 2023, Henderson et al., 2012]. The empirical approach follows standard panel data methods, while explicitly recognizing the constraints imposed by the data.

The treatment variable, D_{it} , is an indicator equal to one if region i is the birth region of the national leader in year t , and zero otherwise [Hodler and Raschky, 2014]. The primary outcome, Y_{it} , is the log of mean annual NTL at the regional level, with regional GDP used as a secondary outcome where data permit.

3.2 Empirical Strategy and Methods

The analysis exploits rare leadership transitions in Ethiopia to examine whether regions associated with the national leader receive systematically higher nighttime light (NTL) intensity. The treatment variable, D_{it} , is an indicator equal to one if region i is the birth region of the national leader in year t , and zero otherwise [Hodler and Raschky, 2014]. The primary outcome, Y_{it} , is the log of mean annual NTL at the regional level; regional GDP is used as a secondary outcome where available.

Between 1992 and 2023, Ethiopia experienced three national leadership transitions (1995, 2012, 2018), affecting a total of four regions. The panel covers 13 regions and 429 region-year observations. Because distributive favoritism may not materialize immediately, the treatment is lagged by one year (D_{it-1}) to allow for policy implementation and data reporting delays. This lag also mitigates mechanical correlation between leadership change and contemporaneous national shocks.

The main estimating equation is:

$$Y_{it} = \alpha + \beta D_{it-1} + \mu_i + \lambda_t + \mathbf{X}_{it}'\gamma + \epsilon_{it} \quad (1)$$

where μ_i and λ_t denote region and year fixed effects, respectively, and $\mathbf{X}_{it}$ is a vector of time-varying controls: log population, log land area, and (in robustness checks) regional conflict events and urbanization rate. All covariates are measured prior to treatment to avoid post-treatment bias.

Identification and Limitations Identification relies on within-region changes in leader status over time. In Ethiopia, this identifying variation is extremely limited: leadership transitions are infrequent, the number of regions is small, and most regions are never treated. The effective sample size for treated observations is therefore very small, and the timing of transitions is highly collinear with national trends. As a result, fixed effects estimates are unstable and sensitive to specification.

Accordingly, the analysis does not claim causal identification. Estimated coefficients are interpreted as descriptive associations, and emphasis is placed on the stability and robustness of results across specifications.

Estimation, Inference, and Diagnostics All models include region and year fixed effects. Standard errors are clustered at the region level to account for serial correlation [Marianne Bertrand and Mullainathan, 2004]. Given the small number of clusters, inference is supplemented with wild cluster bootstrap procedures [Cameron et al., 2008]. Driscoll–Kraay standard errors are also reported to account for cross-sectional dependence [Driscoll and Kraay, 1998]. Placebo tests (randomly assigning leader status to regions) and event-study specifications are used to assess whether estimated effects are distinguishable from noise and to check for pre-trends.

Data The analysis combines three data sources: (i) harmonized nighttime lights data from DMSP-OLS and VIIRS satellites, processed following Li and Gibson [2023]; (ii) regional and national GDP data from the World Bank; and (iii) a hand-coded panel identifying leaders’ birth regions. Administrative boundaries are harmonized to a consistent regional structure; where regions split or merge, values are adjusted using population-weighted aggregation. Missing NTL observations are interpolated within regions. The final dataset consists of 429 region-year observations across 13 regions.

Scope of Inference Findings are most relevant for settings with centralized political authority, limited regional autonomy, and infrequent leadership transitions. In such contexts, the lack of within-unit variation constrains the ability of standard panel methods to identify distributive effects. Null or unstable findings should be interpreted as reflecting design limitations rather than the absence of underlying political processes.

4 Results

4.1 Data and Descriptive Analysis

4.1.1 National NTL–GDP Validation

We begin by validating the use of nighttime lights (NTL) as a proxy for economic activity in the Ethiopian context. Specifically, we regress the natural logarithm of national mean NTL on the natural logarithm of GDP per capita, using “GDP per capita (constant 2015 US\$)” from the World Bank [Bank, 2024]. The estimated elasticity is 1.94 ($p < 0.001$, $R^2 = 0.94$), indicating that a 1% increase in real GDP per capita is associated with a 1.94% increase in mean NTL.

This elasticity is substantially higher than most estimates in the literature. While reported elasticities for developing countries often fall in the range of 0.3 to 1.0 (see, e.g., Henderson et al. 2012 for cross-country evidence and Bickenbach and Bode 2016 for

context-specific results), these ranges are sensitive to country context, time period, and measurement approach. Gibson [2021] highlights that elasticity estimates can vary widely depending on data quality and spatial scale. One plausible explanation for Ethiopia’s high elasticity is rapid electrification and infrastructure expansion over the sample period, including large-scale projects such as the Grand Ethiopian Renaissance Dam [World Bank, 2019]. In such contexts, the expansion of lighting infrastructure may outpace measured increases in output, leading to a stronger elasticity between NTL and GDP.

These results confirm that NTL is a strong proxy for aggregate economic activity at the national level, while also highlighting that its interpretation is context-dependent.

4.1.2 Regional Inequality

We next examine regional disparities using NTL-based Gini coefficients. The coefficient declines from 0.715 in 1992 to 0.513 in 2018, before stabilizing at 0.526 in 2024. This pattern indicates a reduction in regional inequality over time, although substantial disparities persist.

4.1.3 Regional Trends and Distribution

Figure ?? presents NTL trajectories for all regions over the sample period. The data show substantial heterogeneity in both levels and growth rates, with a small number of regions accounting for a large share of overall luminosity.

The distribution of NTL values (Figure ??) is highly skewed, reflecting concentration of economic activity in a limited number of regions. This skewness has implications for estimation, as results may be sensitive to high-luminosity regions.

4.1.4 Leader vs. Non-Leader Regions

Descriptive comparisons between leader and non-leader regions are presented in Figures ?? and ?. While leader regions tend to exhibit higher average NTL levels, these differences are not stable over time and may reflect persistent structural differences rather than changes associated with leadership.

4.2 Regression Results: Leader Birth-Region Effects

We estimate a series of panel regression models to assess the association between leader-region status and regional economic activity. The estimated coefficient on the lagged leader-region indicator is positive across most specifications, but its magnitude and statistical significance vary substantially.

In pooled OLS and random effects models, the coefficient is large and statistically significant. However, once region and year fixed effects are included, the estimated effect declines markedly. In several specifications, the coefficient becomes statistically insignificant, and its magnitude is sensitive to the inclusion of additional controls and sample restrictions.

The between estimator yields imprecise estimates with large standard errors. Across all models, explanatory power remains limited.

Taken together, these results indicate that the estimated relationship between leader-region status and economic outcomes is not stable across specifications.

4.3 Placebo and Event-Study Evidence

To further assess the empirical patterns, we implement placebo and event-study analyses. The event-study estimates show that both pre-treatment and post-treatment coefficients are small and statistically indistinguishable from zero. Confidence intervals are wide and consistently include zero across all periods.

These findings indicate the absence of detectable pre-trends or dynamic treatment effects. However, the limited number of treatment events and the structure of the data constrain the informativeness of these tests.

5 Robustness and Sensitivity Analysis

To evaluate the stability of the results, we implement a series of robustness checks that correspond to key identifying assumptions. Table ?? summarizes these tests.

Across specifications, the estimated leader-region effect is highly sensitive to modeling choices. The coefficient varies substantially across estimators, loses statistical significance under alternative inference procedures, and is not robust to the inclusion of region-specific trends, conflict controls, or sample restrictions such as excluding high-luminosity regions.

Placebo and event-study analyses do not detect systematic pre-trends or treatment effects. While the absence of pre-trends is consistent with identifying assumptions, the lack of detectable treatment effects further underscores the instability of the estimates.

Overall, the robustness analysis does not support the presence of a stable or consistent relationship between leader-region status and regional economic outcomes.

6 Discussion

This study provides a systematic assessment of regional favoritism in Ethiopia using harmonized nighttime lights and GDP data within a panel framework. While some specifications produce statistically significant estimates, these results are not robust across models, inference procedures, or sample definitions.

A central finding is that the absence of consistent results reflects limitations in statistical power and identifying variation rather than definitive evidence against favoritism. Given the sample size and model structure, the analysis is only capable of detecting very large effects—approximately 0.95 log points. In contrast, typical effect sizes reported in the literature range from 0.10 to 0.25 log points, well below the detectable threshold in this setting.

This implies that the analysis cannot rule out effects of a magnitude commonly reported in prior studies. Instead, the results should be interpreted as a diagnostic of the limits of the empirical design.

Two broader insights follow. First, in small-N settings with limited within-unit variation, panel estimators can produce results that are highly sensitive to specification. Apparent statistical significance may reflect model dependence rather than underlying causal relationships. Second, null or unstable findings in such contexts are inherently difficult to interpret without explicit consideration of statistical power.

These findings also inform the interpretation of the broader literature. Existing evidence on regional favoritism often relies on panel designs that may perform differently depending on the availability of variation and the institutional context. The Ethiopian

case illustrates that, in centralized systems with limited regional autonomy and infrequent leadership transitions, the empirical detection of favoritism is particularly challenging.

7 Limitations

The results should be interpreted in light of several limitations. The small number of regions and infrequent leadership transitions severely constrain statistical power and the ability to identify within-region effects. Leader-region assignment is potentially endogenous and may be correlated with unobserved determinants of economic outcomes. Measurement error in NTL and the limited availability of regional GDP data further complicate interpretation. In addition, boundary harmonization may obscure localized variation.

These limitations imply that the analysis cannot provide definitive causal estimates. Instead, the findings are best understood as descriptive patterns conditioned on the available data and empirical design.

8 Policy Implications

The results do not provide evidence that regional favoritism is absent. Rather, they highlight the limits of current data and methods in detecting such patterns. Policymakers should therefore exercise caution in interpreting null findings derived from proxy-based measures such as nighttime lights.

Improving the empirical basis for policy requires investment in more granular and reliable subnational data, as well as institutional arrangements that support consistent measurement across regions. Without such improvements, empirical assessments of distributive outcomes will remain limited.

9 Conclusion

This paper reassesses regional favoritism in Ethiopia using harmonized nighttime lights and GDP data. We find no robust evidence of systematic advantages for leaders' birth regions. However, this result reflects the limited statistical power and identifying variation available in the data rather than a definitive absence of favoritism.

The primary contribution of the paper is to clarify the empirical limits of standard panel approaches in small-N, low-variation settings. The findings demonstrate that, under these conditions, estimated effects are highly sensitive to specification and difficult to interpret.

Future research should prioritize richer data, more granular outcomes, and research designs that provide credible exogenous variation. Transparent reporting of identification constraints and statistical power is essential for advancing empirical work on distributive politics.

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11 Appendix

Summary Statistics for Main Variables

A Supplementary Materials and Diagnostics

A.1 Diagnostic Checklist for NTL-Based Studies

To promote transparency and improve empirical credibility, this study provides a structured diagnostic checklist for research using nighttime lights (NTL) data, particularly in small-N or data-constrained settings.

1. Data Quality and Coverage

Table 1: Summary statistics for NTL, lnNTL, Mean_NTL, and Leader_region (regional panel, 1992–2024).

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
NTL	429	14167.6	25924.6	0.0	1306.0	3359.0	14484.0	169874.0
lnNTL	429	8.16	2.22	-4.61	7.17	8.12	9.58	12.04
Leader_region	429	0.11	0.32	0.0	0.0	0.0	0.0	1.0
Mean_NTL	429	3.10	9.27	0.0	0.02	0.12	1.28	57.16

- Are NTL data harmonized across sensors (e.g., DMSP-OLS and VIIRS)?
- Are administrative boundaries consistent over time?
- Is spatial and temporal coverage appropriate for the research question?

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2. Identification and Variation

- Is there sufficient variation in treatment (e.g., leadership transitions)?
- Are fixed effects included to control for unobserved heterogeneity?
- Is identifying variation robust to alternative sample definitions?

3. Placebo and Falsification Tests

- Are placebo assignments used to benchmark results?
- Are findings distinguishable from random assignment?

4. Power and Sensitivity Analysis

- Is statistical power assessed given the number of treated units?
- Are results robust across specifications and samples?

5. Reporting and Transparency

- Are data and code available for replication?
- Are limitations explicitly acknowledged?

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A.2 Data Construction and Sources

The dataset combines three primary sources:

1. **Nighttime Lights (NTL):** Annual composites from DMSP-OLS (1992–2013) and VIIRS (2012–2024), harmonized following Li and Gibson [2023], Elvidge et al. [2021]. Raster data are aggregated to first-order administrative regions using official shapefiles. Values are log-transformed for analysis.

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2. **Regional GDP:** Regional and national GDP per capita from the World Bank. Where regional GDP is unavailable, NTL is used as a proxy.
3. **Leader-Region Indicator:** Hand-coded variable equal to one if a region corresponds to the national leader’s birthplace in a given year, following Hodler and Raschky [2014]. Coding protocols are documented in the supplementary materials.

Administrative boundaries are harmonized to a consistent regional structure. Where regions split or merge, values are adjusted using population-weighted aggregation. Missing NTL observations are interpolated within regions.

The final dataset contains 429 region-year observations across 13 regions.

A.3 Identification Threats and Diagnostics

Threat	Description	Diagnostic Approach
Omitted Variables	Unobserved shocks correlated with treatment and outcome	Region and year fixed effects, placebo tests
Reverse Causality	Economic conditions influence leader emergence	Lagged treatment, placebo tests
Measurement Error	NTL imperfect proxy for economic activity	Use of harmonized datasets, robustness checks
Selection Bias	Leader regions differ systematically	Fixed effects, pre-trend analysis
Bad Controls	Inclusion of endogenous controls	Restrict to pre-determined controls

Table 2: Identification threats and diagnostic approaches. These diagnostics are informative but do not fully resolve identification concerns.

A.4 Extended Identification Challenges in Small-N Panels

Empirical identification of political favoritism is particularly challenging in small-N panel settings such as Ethiopia. Three key issues arise.

Omitted Variable Bias Without time fixed effects, national shocks correlated with leadership transitions may bias estimates. Apparent favoritism effects may reflect macroeconomic fluctuations rather than causal relationships.

Limited Identifying Variation Including time fixed effects absorbs common shocks but reduces identifying variation. When treatment timing is highly collinear with time effects, models cannot distinguish treatment effects from national trends.

Low Statistical Power The small number of regions and limited treatment variation constrain statistical power. Estimates are imprecise, and placebo tests often yield similar distributions to observed results.

Implication Taken together, these challenges mean that standard panel methods, when applied to small-N settings like Ethiopia, are fundamentally limited in their ability to detect or rule out moderate favoritism effects. As a result, null or unstable findings in this context do not provide evidence against the existence of favoritism, but rather highlight the need for richer data, more exogenous variation, or alternative research designs. Researchers and policymakers should therefore be wary of drawing strong substantive conclusions from null results in data-constrained environments, and should prioritize transparency about these limitations in future work.

In settings like Ethiopia, where data and institutional constraints sharply limit statistical power, null results from standard panel methods do not rule out moderate regional favoritism; instead, they reveal the fragility of empirical inference and the need for richer data or alternative research designs to credibly detect distributive politics.

A.5 Reproducibility and Transparency

All data processing and estimation procedures are fully documented and reproducible.

- Data sources, cleaning procedures, and variable construction are documented in the supplementary materials.
- Code for data processing and estimation is available in a public repository.
- The full workflow—from raw data to final results—is version-controlled and reproducible.
- All sample restrictions, transformations, and assumptions are explicitly documented.

Researchers can replicate all results by following the instructions provided in the project repository.